1

Good morning. I am Mhamad Naser Dine, and this is Ali Srour. Look at this motor and its current: a clean sine wave. This motor looks healthy. Is it? Keep this question in mind. We will answer it live at the end.

2

Induction motors run pumps, fans and conveyors in every factory. When one stops, the production stops. And motors break slowly. Most faults come from three places. Rotor bars crack and break. Stator windings: the insulation fails and turns short-circuit. Bearings wear out. These faults start small and grow over weeks. So our goal: detect the fault while it is still small.

3

Here we broke 3 rotor bars: the orange bar. Teal is the current of the healthy motor. Dashed orange is the motor with 3 broken bars. The two currents are almost identical: the eye cannot see the fault.

4

But it is clearly visible in the frequency spectrum. This video shows the same motor while the fault grows. Healthy motor: one peak at 50 hertz. With 1 broken bar, two side peaks appear, at $(1 \pm 2s)\cdot f$, exactly where the theory says. With 3 broken bars, they get even stronger. The fault grows, and its fingerprint grows with it. This is called motor current signature analysis.

5

Each fault leaves a different fingerprint. Broken bars: side peaks. Bearing defect: peaks linked to the bearing. Inter-turn short: the Park circle becomes an ellipse. Supply unbalance: the voltage itself is unbalanced. The network must learn all four patterns, plus the healthy one.

6

A network cannot read thousands of samples. So each motor becomes 19 numbers: time, balance, Park, spectrum and slip. For our faulty motor, the orange broken-bar sidebands are the highest. Ali will now show how the network learns.

7

Thank you, Mhamad. The 19 numbers become the 19 inputs of a neural network, with 15 hidden neurons and 5 outputs, one per condition. The hidden layer combines the inputs. The highest output is the diagnosis: Broken bars, 96 %. It learns with back-propagation: the error goes backward and corrects every weight. After many examples, its error becomes very small.

8

To learn, it needs examples. The network learns from 1500 virtual motors, 300 per condition, with different loads and noise. One dot = 10 simulated motors.

9

We split them, exactly as in the course. Training, 60 %: it adjusts the weights. Test, 20 %: it stops at point A. Validation, 20 %: never seen during learning. It is the final, honest grade. Every number we show comes from this set.

10

The training error keeps falling. But the test error stops falling at A, epoch 149. Point A: we keep these weights, before the network starts memorising.

11

Why do we need a hidden layer? Here we keep only two features. Healthy motors sit in one corner. A single layer can only draw a straight line: Perceptron, 86.7 %.

12

A hidden layer bends the boundary around the corner: 92.5 %. That is why we use a multilayer perceptron.

13

We compared seven networks, all written from scratch, each trained ten times. MLP-BP is the best: 94 %. Momentum and adaptive η are almost the same. MLP-LM is the fastest. ADALINE gets 90 %, and the RBF only 78 %: its Gaussian neurons only see a small region, so they cannot cover 19 dimensions.

14

We tested on 500 new motors never seen. Even one broken bar is found 89 % of the time. 100 % of medium and severe faults. 8 hidden neurons are enough. Our limit: at 5 % sensor noise, accuracy drops to 31 %. Now, back to our mystery motor. Mhamad.

15

Back to our question: is the mystery motor healthy? The program creates a motor and hides its condition. The network reads only the current. We reveal the truth. Even we do not know the condition. The verdict: faulty, broken rotor bars, 96 %. The bars show how sure it is. The network was right. Let’s try one more. Now the hardest case: one single broken bar, at 40 % load. Even one broken bar is detected. Look at the small side peaks. From the current to the diagnosis in one second. Ali will conclude.

16

To conclude: from the current to the diagnosis. 94 % with MLP-BP, 89 % of early faults found, and an honest evaluation: 3 sets, point A, 10 runs. Next: real motor data, the severity, and the remaining life. Thank you. Any questions?